

Instructions: Please answer the following questions making sure to write your answers legibly on the second page. The code below is a segment of a Python application that provides context for the questions below.

- Since we are writing Python code, please be careful about spacing/indentation. Leave enough room that it is obvious when you want things indented.
- Make sure to write legibly.
- All questions are worth equal points unless otherwise stated.
- The quiz is cumulative and may include material from previous weeks.

process.py

```
from data_utils import (
    process_data,
    validate_input,
    transform_output
)

def calculate_sum(a, b):
    return a + b

def calculate_product(a, b):
    return a * b

def square(x):
    return x * x

def double(x):
    return x * 2

def increment(x):
    return x + 1

if __name__ == '__main__':
    result = calculate_sum(10, 5)
    print(f"Result: {result}")
```

1. Look at the `calculate_sum` function above. This function should work with all numeric types. Please rewrite it so that when you enter `print(calculate_sum.__doc__)` in Python, it returns the text: `Adds two numbers together`.

2. Please write a function called `apply_to_all` that:

- Takes two arguments: `func_list` (a list of functions) and `x` (a value)
- Applies each function in `func_list` to `x`
- Returns a list of results from applying each function to `x`

For example: `apply_to_all([square, double, increment], 5)` should return `[25, 10, 6]`.

3. Please write a function called `flex_print` that accepts any number and type of keyword arguments (but no positional arguments) and prints each key-value pair in the format: "Key: {key}, Value: {value}" (one per line). If the value is an integer, instead print: "Key: {key}, Value: Integer".

4. [4 Points] In class we discussed how code complexity can be broken down into two categories: *essential* complexity and *accidental* complexity. Please list two sources of accidental code complexity.